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Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs

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Code, tests & data

Estimate position bias from ordinary click logs — weight each repeated query–item pair by $\omega_i = \frac{N_k^i N_{k'}^i}{N_k^i + N_{k'}^i}$ (harmonic; N = pair's impressions at each rank).

On the full KDD log, equal weighting needs ≈ 4.5× **more impressions** for the same estimated precision.

(The 4.5× is model-based precision on KDD's observed counts; ground-truth simulations in Block 6 corroborate it.)

1. Same item, different rank ⇒ position bias

Users click top results more — partly because they are *better*, partly just because they are *seen* more. Under the **Position-Based Model**, a click factorises as

$$P(\text{click} \mid q, d, k) = \underbrace{p_k}_{\substack{\text{examination:}\\ \text{seen at rank } k}} \times \underbrace{r_{q,d}}_{\text{relevance}}.$$

The **same product X**, shown at **different ranks** across groups of logged sessions, lets us cancel relevance:

Sessions: X at rank 1

Query: red shoes
1. Product X
2. other item
3. other item

1,000 impressions → 100 clicks → **10% CTR**

Sessions: X at rank 3

Query: red shoes
1. other item
2. other item
3. Product X

1,000 impressions → 40 clicks → **4% CTR**

Same query & item ⇒ same r, and $\text{CTR}_k = p_k r$:

$$\frac{\text{CTR@1}}{\text{CTR@3}} = \frac{10\%}{4\%} = \frac{p_1 r}{p_3 r} = \frac{p_1}{p_3} = 2.5\times$$

The CTR ratio *is* the position-bias ratio — no A/B test, no relevance labels. Reusing these natural rank changes is called *intervention harvesting*; the task is to combine **millions** of such query–item cells into one estimate of $R_{k,k'} = p_k / p_{k'}$.

Key terms: *cell* i = one query–item pair · N_k^i = times it was shown at rank k · C_k^i = clicks there · ω_i = its weight.

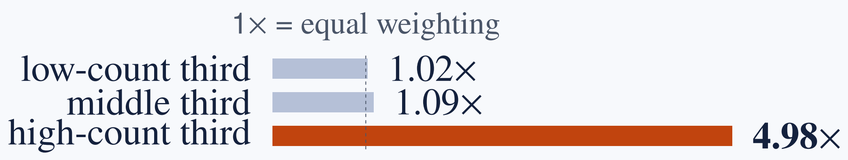
2. Why equal weighting fails on real logs

On the **full KDD Cup 2012** log (235.6M impressions), the cells seen at both rank 1 and rank 2 — **4.61M** query–item pairs, 132.3M impressions — have wildly uneven counts:

60.7%
of pairs are near-empty ($N_1 + N_2 \le 5$)

42%
of all impressions sit in the top **0.1%** of pairs

Giving a near-empty cell the **same say** as a million-impression cell averages noisy guesses and precise measurements equally. The gain from fixing this lives in the high-count head:



Efficiency ($\text{Var}_{\text{equal}}/\text{Var}_{\text{harmonic}}$) by cell-count tertile on KDD.

Why not Baidu-ULTR? In its public release 94.1% of (q, d, pos) cells have N=1 — consistent with per-cell deduplication, unlike KDD's heavy tail.

3. Drop-in fix: weighted ratio estimator

Under PBM, *any* count-only weight $\omega_i(N_k^i, N_{k'}^i)$ in the self-normalised ratio estimator still targets the same $p_k/p_{k'}$ — it changes only the noise; harmonic is the one that minimises the count-only variance surrogate:

$$\hat{R}_{k,k'} = \frac{\sum_i \omega_i \left(C_k^i / N_k^i\right)}{\sum_i \omega_i \left(C_{k'}^i / N_{k'}^i\right)} \quad \text{with} \quad \omega_i = 1 \longrightarrow \omega_i = \frac{N_k^i N_{k'}^i}{N_k^i + N_{k'}^i}$$

Scheme	Weight ω_i	Property
Equal *(Agarwal et al.)*	1	Treats N=1 and N=10^6 equally
Min-count *(new)*	$\min(N_k^i, N_{k'}^i)$	Heuristic; no all-count guarantee
Harmonic *(new)*	$\frac{N_k^i N_{k'}^i}{N_k^i + N_{k'}^i}$	**Count-only surrogate IVW optimum**

One line of code. Composes cleanly with AdjacentChain, AllPairs, or Pivot — no labels, no tuning.

4. KDD Cup 2012 results: ranks 1 vs 2 (4.61M pairs)

Scheme	$\hat{R}_{1,2}$	SE (lower better)	**Efficiency vs equal**
Equal ($\omega_i = 1$)	1.683	4.24e-3	1.00
Min-count	1.712	2.05e-3	4.41
Harmonic	**1.710**	**2.02e-3**	**4.53**

4.53×
impressions needed by equal weighting to match harmonic

✓ **Ranks (2,3):** $\hat{R}_{2,3}$=1.73, **3.14× efficiency vs equal** (2.1M pairs).

✓ **Cap-robust:** efficiency stays **4.12×** at the 99.99th-percentile cap and **1.86×** at the 90th.

✓ **Calibrated:** Plug-in SE matches click-resample SE within **1–2%**.

Same answer, far less data. Equal and harmonic weighting land within ~2% of each other (1.68 vs 1.71) — harmonic simply gets there with ~4.5× fewer impressions. (Under PBM misspecification each weight defines a slightly different average; the gain is precision, not a different "truth".)

5. Theoretical guarantees (why harmonic?)

✓ **Surrogate IVW optimum (Thm. 2):** Replacing unknown cell relevances r_i with a common factor yields the count-only variance surrogate $V_k(\omega) + V_{k'}(\omega)$, which harmonic minimises.

✓ **Unconditional Cauchy–Schwarz bound (Thm. 3):** $V_k(\omega^h) + V_{k'}(\omega^h) \le V_k(\mathbf{1}) + V_{k'}(\mathbf{1})$ for *any* counts $\{N_k^i, N_{k'}^i\}$ (min-count has no such guarantee). On KDD's asymmetric surrogate $\alpha V_k + \beta V_{k'}$, harmonic still improves **4.75×**.

✓ **Why not pair-bootstrap?** Resampling (q, d) pairs perturbs the few high-information cells and entangles efficiency with model fit; the closed-form (delta-method) $\widetilde{V}$ isolates conditional efficiency.

6. Ground-truth validation & PBM guardrail

Independent ground-truth simulations confirm the delta-method efficiency gains and test robustness under explicit click-model violations:

up to 54%
Synthetic PBM
less variance vs equal; up to **2× data savings**

~40%
Yahoo-LTR
less variance (CI: 29–51%); clipping inflates MSE **458×**

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PBM Violations
lower MSE than **Equal** under **Trust**, **Cascade** & **Pos-Dep**

Guardrail (Cochran's Q). Across 16,132 high-information KDD pairs (≥ 5 clicks/rank), per-cell odds ratios vary 3.83× more than binomial click noise (Q/df = 3.83), confirming PBM is an approximation and weight choice is a precision vs. estimand trade-off.

Bottom line

Replace $\omega_i = 1$ with $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$: **one line of code**, no extra data, ~4.5× higher efficiency on full KDD (ranks 1 vs 2), and lower MSE than equal weighting under all three tested PBM violations — no relevance labels, no tuning.

github.com/alemagnani/surrogate-ivw-intervention-harvesting • DOI 10.1145/3799682.3839883
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